

Ishaan Gupta

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EDUCATION

UNIVERSITY OF CALIFORNIA, BERKELEY

Graduation: May 2027

- M.S. Mechanical Engineering, Concentration in Control Systems
- B.S. Mechanical Engineering, Minor in Electrical Engineering & Computer Science (EECS)
- **Relevant Coursework:** Power Electronics Design, Design of Microprocessor-Based Mechanical Systems, Dynamic Systems and Feedback, Microelectronic Devices and Circuits, Introduction to Digital Design and ICs (In Progress)

WORK EXPERIENCE

Graduate Student Researcher

May 2026 - Present

Boles Group, Berkeley Power and Energy Center | Berkeley, CA

- Exploratory work on the use of piezoelectric materials in replacing passive components in power conversion
- Developing burst-mode control for a piezoelectric power converter, synthesizing novel switching sequences for high-efficiency, low-power conversion, and modeling system dynamics in PLECS and MATLAB

Hardware Engineering Intern

May 2025 - Aug 2025, May 2024 - Dec 2024

Neptune Medical | Burlingame, CA

- Aided R&D efforts for a deep endoluminal flexible robot; assembled and tested custom brushless motors, designed manufacturing fixtures, performed tolerance stack analysis, and used GD&T to draft manufacturing drawings
- Designed a double redundant instrument detection sensor assembly with micrometer precision over a wide thermal range. Responsible for determining system requirements, sensor selection, and initial validation with a quick-turn prototype.
- Designed automated fluidics infrastructure for endurance testing flexible robotics assemblies, designed and integrated PLC controllers and power electronics into a sealed electric cabinet, and wrote firmware in C++

Hardware Engineering Intern

Feb 2024 - May 2024

Covariant.ai | Emeryville, CA

- Ran high-stakes investor demos to showcase the capabilities of pick-and-place robots, including NVIDIA and Amazon
- Assembled electro-pneumatic components and conducted end-of-line testing on vacuum generator assemblies
- Designed filters for end-of-arm gripper tooling to reduce plastic particulate build-up from customer product packaging

Student Research Assistant

Jun 2023 - May 2024

A-Lab, Ceder Group, Lawrence Berkeley National Laboratory | Berkeley, CA

- Aided in developing an autonomous laboratory capable of synthesizing materials for battery applications, utilizing robotics and artificial intelligence. Worked alongside doctoral students and postdocs under Professor Gerbrand Ceder
- Improved a robot cell handling sample cap dispensing and cap placing to reduce failure rate from 2% to .01%.
- Fully owned the design, fabrication, and testing of a cell capable of automatically capping and decapping cap samples by repurposing an Ender 3 3-D printer with custom firmware and pneumatic SMC robotics grippers for rapid deployment and integration. Designed, built, and tested a lab safety-compliant 240-volt electrical and pneumatic box.

PROJECTS

DC-DC ZVS Resonant-Transition Synchronous Buck Converter w/ MPPT

Jan 2026 - May 2026

Power Electronics Design | University of California, Berkeley

- Designed a 100W DC-DC Zero Voltage Switching (ZVS) Resonant-Transition Synchronous Buck Converter, utilizing PLECS to simulate power stages and regulate a 16-24V solar panel input to efficiently charge a 12V battery.
- Engineered custom AC magnetics using a 3C95 RM8 ferrite core, designed an optimized PCB layout in Altium to minimize high-frequency parasitic inductance, and programmed an adaptive Perturb & Observe MPPT algorithm in C++.
- Experimentally verified soft-switching operation across varying load conditions, achieving a final hardware efficiency of 88.7%, a dynamic MPPT score of 70.94W, and a high power density of 2.825 W/cm³.

V/STOL Drone

Aug 2025 - Dec 2025

Mechatronics Design | University of California, Berkeley

- Designed a custom flight computer using an ESP32 MCU and a 9-axis MPU-9250 IMU for a tilt-rotor V/STOL RC plane
- Modeled system dynamics and designed a PID controller with a Mahony filter for 3-axis hover stabilization in C++

Drawing Robot

Jan 2025 - May 2025

Design of Microprocessor-Based Mechanical Systems | University of California, Berkeley

- Designed a custom stepper motor controller PCB around the RP2040 for a robotic pen plotter. Implemented field-oriented control in Raspberry Pi SDK using C++, implemented an RTOS, and CAN communication with an ESP32

SKILLS

- **Engineering Skills:** CAD/CAM, FEA, Rapid Prototyping, Product Design, GD&T, Design for Manufacturing, Microcontroller Programming, Embedded Systems, PCB Layout, Circuit Design, Circuit Validation, Circuit Simulation
- **Fabrication Skills:** CNC Manufacturing, Manual Mills, Hand Tools, 3-D Printing, Waterjet and Laser Cutting, Soldering
- **Programming:** Java, Python, C++, MATLAB, LabVIEW, ESP-IDF, Arduino
- **Software:** SolidWorks (CAD/CAM/Simulation), OnShape, Fusion 360, Altium, KiCad, PLECS, LTSpice
- **Certifications:** CSWP - Certified SolidWorks Professional